

Why do we need a software test suite?

Software test suites provide a lot of benefits:

- It is a very cost-effective way of testing a system under test (SUT) and finding defects or errors in the development stages, i.e. at the early stages.
- Software test suites are much more reliable and error-free than manual testing. So customer reliability in an organization is enhanced, and customer satisfaction is better ensured.
- Software test suites provide better quality products.
- Software tested products are of high quality and hence show fewer defects, reducing the maintenance cost of the product.

Properties of a software test suite

There are many features or properties of a software test suite:

- A properly elaborated test plan is created and based on that, a relevant test suite is formulated.
- A test suite mentions the purpose of all the tests included for testing the system under test (SUT).
- Test suites can be formed based on either a test scope or a test cycle.
- It has all the test parameters like environment, application, etc. for each particular test.
- A large number of tests and test cases needed for the system under test (SUT) is elaborately mentioned.
- Test suites provide ways for speeding up tests to be done on the system under test (SUT) and hence increases the performance of testing.
- Test suites include non-functional as well as functional tests.
- Test suites can be well utilized for automated tools like JUnit, Selenium, etc.

What are the varieties of a software test suite?

There are two varieties of a software test suite:

- **Abstract Test Suite:** A high-level model is created for every system before testing it. A cluster of abstract test cases from this high-level model is called an abstract test suite. The abstract test suite is a piece of the high-level model created from the system under test (SUT). These are based on a high-level model, so many ground details about the software can be missing. So an abstract test suite cannot be performed directly on software.
- **Executable Test Suite:** Another type of test suite is an executable test suite that can be performed directly on software. It is extracted from the abstract test cases. It provides the ground details needed to execute a test suite. The